

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Dot .Net Core and Progressive Web Application Practical

Practical - IX

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Class: TYIT	Batch: 1
Date of Assignment: 18/08/2026	Date/Time of Submission: 18/08/2026

a. Develop CRUD operations using ASP.NET Core Web API and Entity Framework Core.

Code: StudentController.cs:

```
202608180310...tialCreate.cs StudentController.cs Program.cs appsettings.json
StudentAPIWithEF StudentAPI.Controllers.StudentsContro

1 using Microsoft.AspNetCore.Mvc;
2 using StudentAPI.Data;
3 using StudentAPI.Models;
4
5 namespace StudentAPI.Controllers
6 {
7     [Route("api/[controller]")]
8     [ApiController]
9     public class StudentsController : ControllerBase
10    {
11        private readonly ApplicationDbContext _context;
12
13        public StudentsController(ApplicationDbContext context)
14        {
15            _context = context;
16        }
17
18        // GET ALL
19        [HttpGet]
20        public IActionResult GetStudents()
21        {
22            return Ok(_context.Students.ToList());
23        }
24
25        // GET BY ID
26        [HttpGet("{id}")]
27        public IActionResult GetStudent(int id)
28        {
29            var student = _context.Students.Find(id);
30
31            if (student == null)
32                return NotFound();
33
34            return Ok(student);
35        }
36    }
37 }
```

Program.cs:

```
202608180310...tialCreate.cs StudentController.cs* Program.cs* appsettings.json
StudentAPIWithEF
1 using Microsoft.EntityFrameworkCore;
2 using StudentAPI.Data;
3
4 var builder = WebApplication.CreateBuilder(args);
5
6 builder.Services.AddControllers();
7
8 builder.Services.AddDbContext<AppDbContext>(options =>
9     options.UseSqlServer(
10         builder.Configuration.GetConnectionString("DefaultConnection")));
11
12 builder.Services.AddEndpointsApiExplorer();
13 builder.Services.AddSwaggerGen();
14
15 var app = builder.Build();
16
17 app.UseSwagger();
18 app.UseSwaggerUI();
19
20 app.UseHttpsRedirection();
21 app.UseAuthorization();
22 app.MapControllers();
23
24 app.Run();
```

appsetting.json:

```
Schema: https://www.schemastore.org/appsettings.json
1 {
2     "ConnectionStrings": {
3         "DefaultConnection": "Server=(localdb)\\MSSQLLocalDB;Database=StudentDB;Trusted_Connection=True;"
4     }
5 }
```

AppDbContext.cs:

```
StudentAPIWithEF StudentAPI.Data.AppDbContext
1 using Microsoft.EntityFrameworkCore;
2 using StudentAPI.Models;
3
4 namespace StudentAPI.Data
5 {
6     public class AppDbContext : DbContext
7     {
8         public AppDbContext(DbContextOptions<AppDbContext> options)
9             : base(options)
10         {
11         }
12
13         public DbSet<Student> Students { get; set; }
14     }
15 }
```

Student.cs:

```
StudentAPIWithEF StudentAPI.Models.Student
1 namespace StudentAPI.Models
2 {
3     3 references
4     public class Student
5     {
6         0 references
7         public int Id { get; set; }
8
9         2 references
10        public string Name { get; set; }
11
12        2 references
13        public int Age { get; set; }
14
15        2 references
16        public string Course { get; set; }
17    }
18 }
```

Output:

Get:

Code	Details
200	Response body<pre>[{ "id": 1, "name": "Sachin", "age": 21, "course": "BCA" }]</pre> Response headers<pre>content-type: application/json; charset=utf-8 date: Tue, 18 Aug 2026 04:54:19 GMT server: Microsoft-IIS/10.0 x-powered-by: ASP.NET</pre>
Responses	
Code	Description
200	OK

Post:

```
{
  "name": "Sachin",
  "age": 21,
  "course": "IT"
}
```

Code	Details
200	Response body<pre>{ "id": 2, "name": "Sachin", "age": 21, "course": "IT" }</pre> Response headers<pre>content-type: application/json; charset=utf-8 date: Tue, 18 Aug 2026 04:55:59 GMT server: Microsoft-IIS/10.0 x-powered-by: ASP.NET</pre>
Responses	
Code	Description
200	OK

GET /api/Students/{id}

Parameters

Name	Description
id * required	
integer(\$int32)	2
(path)	

Execute

Code	Details				
200	Response body <pre>{ "id": 2, "name": "Sachin", "age": 21, "course": "IT" }</pre> Response headers <pre>content-type: application/json; charset=utf-8 date: Tue, 18 Aug 2026 04:58:45 GMT server: Microsoft-IIS/10.0 x-powered-by: ASP.NET</pre> Responses <table> <tr> <th>Code</th><th>Description</th></tr> <tr> <td>200</td><td>OK</td></tr> </table>	Code	Description	200	OK
Code	Description				
200	OK				

id * required

integer(\$int32)
(path)

Request body

Edit Value | Schema

```
{
  "id": 2,
  "name": "Sachin Mehta",
  "age": 20,
  "course": "BScIT"
}
```

Code	Details
200	Response body <pre>{ "id": 2, "name": "Sachin Mehta", "age": 20, "course": "BScIT" }</pre> Response headers <pre>content-type: application/json; charset=utf-8 date: Tue, 18 Aug 2026 05:00:39 GMT server: Microsoft-IIS/10.0 x-powered-by: ASP.NET</pre>
Responses	
Code	Description
200	OK

Name	Description
id * required integer(\$int32) (path)	2
Execute	
Responses	

Code	Details
200	Response headers <pre>date: Tue, 18 Aug 2026 05:01:32 GMT server: Microsoft-IIS/10.0 x-powered-by: ASP.NET</pre>
Responses	
Code	Description
200	OK

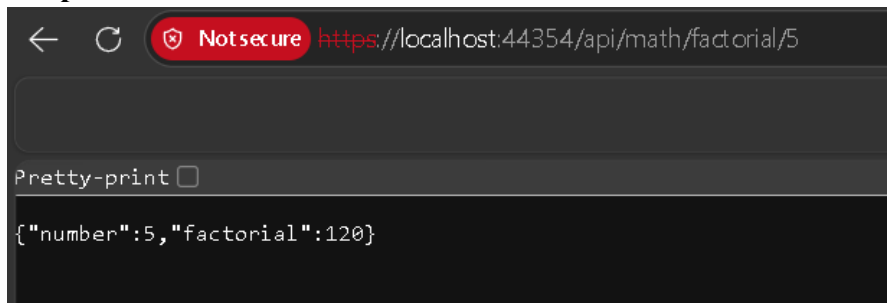
b. Create an API endpoint to calculate Factorial of a Number and test it using Postman.
Code:

```

using Microsoft.AspNetCore.Mvc;
namespace FactorialAPI.Controllers
{
    [Route("api/[controller]")]
    [ApiController]
    public class MathController : ControllerBase
    {
        [HttpGet("factorial/{number}")]
        public IActionResult GetFactorial(int number)
        {
            if (number < 0)
            {
                return BadRequest(
                    "Factorial not possible for negative numbers.");
            }
            long factorial = 1;
            for (int i = 1; i <= number; i++)
            {
                factorial *= i;
            }
            return Ok(new
            {
                Number = number,
                Factorial = factorial
            });
        }
    }
}

```

Output:



My Collection / Get data Save Share

GET https://localhost:44354/api/math/factorial/5 Send

Docs Params Authorization Headers (6) Body Scripts Settings Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body Cookies Headers (5) Test Results (1/1) ↺

{ } JSON Preview Visualize

```
1 {  
2   "number": 5,  
3   "factorial": 120  
4 }
```

My Collection / Get data Save Share

GET https://localhost:44354/api/math/factorial/-2 Send

Docs Params Authorization Headers (6) Body Scripts Settings Cookies

Query Params

	Key	Value	Description	...	Bulk Edit
	Key	Value	Description		

Body Cookies Headers (5) Test Results (0/1) ↺

Raw Preview Debug with AI

```
1 Factorial not possible for negative numbers.
```